

Frequency Response Part 2

EGR 345

GVSU

It was the best of lectures, it was the worst of lectures....

- I love talking about frequency response and Bode
- I need to working on breaking my lectures into chunks
 - more examples
 - planning on where to break things between lectures

Key Points from Last Lecture

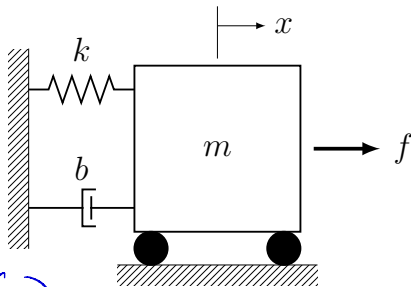
- The transfer function does not change based on the input
 - yes, we solve for $G(s) = Out(s)/In(s)$, but we assume $In(s)$ is unknown
- frequency response (Bode) only applies to stable systems
 - transients die out

What do we mean by frequency response?



- $u(t) = \sin(\omega t)$ ← sinusoidal input
 - $y_{ss}(t) = M \sin(\omega t + \phi)$ ← output is a sine wave at the same frequency
 - assuming $G(s)$ is stable
 - transient terms die out
- Steady state → after transients die out
- magnitude and phase depend on input frequency₃

Applying this to SDOF MSD



$$Y(s) = G(s) \cdot U(s)$$

$$G(s) = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

and

$$U(s) = \frac{\omega}{s^2 + \omega^2}$$

$$Y(s) = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{\omega}{s^2 + \omega^2}$$

~~just use
power point~~

PFE

rows 6+7 rows 8+9

$$D \left(Y(s) = \frac{\frac{1/m \cdot \omega}{s^2 + 2\zeta\omega_n s + \omega_n^2}}{s^2 + \omega^2} = \frac{C_1\omega + C_2s}{s^2 + \omega^2} + \frac{C_3\omega_d + C_4(s+a)}{(s+a)^2 + \omega_d^2} \right)$$

D

$$\frac{1}{m} \cdot \omega = (C_1\omega + C_2s)(s^2 + 2\zeta\omega_n s + \omega_n^2) + (C_3\omega_d + C_4(s+a))(s^2 + \omega^2)$$

as $t \rightarrow \infty$, assuming $\omega \rightarrow 0$

$$y(t) = C_1 \sin \omega t + C_2 \cos \omega t + e^{-\alpha t} \underbrace{(C_3 \sin \omega_d t + C_4 \cos \omega_d t)}_{\text{transient terms}}$$

$$y_{ss}(t) = C_1 \sin(\omega t) + C_2 \cos(\omega t)$$

$$\frac{1}{n} \omega = (C_1 \omega + C_2 s) (s^2 + 2\zeta \omega_n s + \omega_n^2) + (C_3 \omega_d + C_4 (s + \sigma)) (s^2 + \omega^2)$$

$$\frac{1}{n} \omega = (C_1 \omega + C_2 j \omega) (-\omega^2 + 2\zeta \omega_n j \omega + \omega_n^2)$$

$$\frac{1}{n} \frac{\omega}{-\omega^2 + 2\zeta \omega_n j \omega + \omega_n^2} = C_1 \omega + C_2 j \omega$$

$\zeta(s = j\omega) \leftarrow$ bonus / foreshadowing

$$s = j\omega$$

$$s^2 = (j\omega)^2$$

$$s^2 = j^2 \omega^2$$

$$s^2 = -\omega^2$$

$$\sqrt{-1} = i = j$$